

AA-05

Admin Portal Specification

Forty-one screens for internal staff, plus the agronomist Advisor console

30 August 2026 · Rev 1.0 — baseline

SCREENS	PERMISSIONS	SYSTEM ROLES
41	56	9
STEP-UP GATED	ADMIN ENDPOINTS	POLL INTERVAL
17	93	20 s

This document specifies the internal staff console at admin.atavishaala.in: every screen, what gates it, what an operator can and cannot do on it, and the four states each one has to implement. It is written for FE-1, who builds it, and for whoever reviews the pull requests. It assumes AA-00 section 0.2 for why a separate portal exists at all, AA-04 for the endpoint behind every action, AA-08 for the permission and PII rules, AA-01 section 1.5 for the kit components named throughout, and AA-11 for the week each screen lands in.

5.1 WHAT THE ADMIN PORTAL IS FOR

An exception console for a marketplace that mostly runs itself

The admin portal exists because a three-sided marketplace generates exceptions that no state machine can close on its own. A delivery job goes through three offer waves and nobody accepts it. A rider marks a delivery complete from 340 metres away when `POD_GEOFENCE_METRES` is 200. A fleet declares ₹18 400 of collected cash and remits ₹17 900. A farmer uploads three photos of a diseased leaf and the two-hour advisory clock starts running. Each of those is a decision a person has to make, and the portal is the place where that person sees the work and the evidence for it in the same view.

That framing is a design constraint, not a slogan. Every screen in the inventory in section 5.3 has to answer one of two questions: what needs a decision right now, or what is the evidence behind the decision I am about to make. A screen that answers neither is a screen we do not build in these ten weeks. The test is deliberately blunt because the alternative — a portal that grows one screen per table because the table exists — is how internal tools become unmaintainable by week six.

Two things follow from that, and both are worth stating as prohibitions because they will be argued about. First, this is not a reporting tool. Reporting is the eleven fixed tiles on AD-04, computed by one SQL query and cached 30 s at the edge, plus a masked CSV export capped at `EXPORT_ROW_CAP` = 5000 rows. There is no chart builder, no pivot, no date-range explorer. A hosted BI layer is named as a paid gap in AA-00 section 0.9 at roughly \$85/month for Metabase Cloud; until someone signs that cheque, an analytical question is answered by exporting the CSV and opening a spreadsheet. That is genuinely worse than a BI tool and we are choosing it on price.

Second, this is not a data editor. There is no generic row editor anywhere in the portal, no "edit JSON" affordance, and no screen that lets an operator set a status field to a value of their choosing. Mutations happen through named actions that map one-to-one onto the endpoints in AA-04 — `/admin/jobs/:id/rebroadcast`, `/admin/refunds/:id/approve`, `/admin/shipments/:id/transition` — each carrying the permission that gates it and each writing an `audit_log` row with a before and after diff under invariant I-08. An operator can do a lot of damage through those actions; what they cannot do is damage that leaves no trace of who did it and what it looked like before.

THE PREMISE IN ONE LINE

If a screen does not either surface work that needs a human decision or supply the evidence for one, it does not ship in the ten weeks. Ask that question in the pull request.

WHY A SEPARATE PORTAL AT ALL

AA-00 section 0.2 has the full argument: a distinct origin, a distinct token audience (`atv_admin`), a 7-day refresh window against 30 for the app, and mandatory TOTP. None of that can be retrofitted onto a shared front end without weakening it.

5.2 ROLES AND WHAT EACH ONE SEES

9 system roles, 56 permissions, 17 of them step-up gated

There are 9 system roles and no role editor. Cut C-4 in AA-00 section 0.8 removes the custom-role screen and saves four dev-days; the role.write permission exists in the matrix so the endpoint contract does not change later, but no screen in this document calls it. If a seat does not fit one of the nine, the answer for ten weeks is a second role assignment, not a new role.

ROLE	WHAT IT GRANTS	WHO HOLDS IT
SUPER_ADMIN	Everything except raw PII, which no role grants	Founder and technical lead only. Two holders maximum, enforced by a check in staff.manage.
OPS_MANAGER	All Orders, Delivery, Advisory read; job.assign, job.rebroadcast, order.cancel, order.return.review	Day-to-day exception handling. The busiest role.
OPS_AGENT	order.read, order.note.write, job.read, customer.read, advisory.read	Read-mostly support seat. Cannot cancel, refund or reassign.
CATALOG_MANAGER	All Catalog including moderate and price.override; partner.read	Owns the shelf. The only role that approves partner listings.
FINANCE	All Money and Payouts; partner.bank.verify, fleet read	Refund approval and the weekly release. Cannot touch catalog or orders.
PARTNER_MANAGER	All Partners; catalog.read, payout.read	Supplier relationship owner: onboarding, KYC, commission, suspension.
FLEET_MANAGER	All Fleets and Delivery; payout.read	Delivery relationship owner: onboarding, tiers, service areas, rider suspension.
AGRONOMIST	All Advisory except sla.override; customer.read	The expert console seat. Sees crop media and the masked caller ID, nothing else.
AUDITOR	Every *.read permission and audit.read. No write permission at all.	For a reviewer or an external accountant. Deliberately cannot export raw anything.

A staff user can hold more than one role and the effective permission set is the union of the ALLOW entries minus any DENY entry. DENY wins unconditionally, including over SUPER_ADMIN, which is how a suspended finance seat is neutralised in one write without unpicking role membership. GET /admin/me returns the flattened effective permission array, and that array — not the role names — is what the front end holds.

Navigation is built from that array. The kit's Nav component takes the permission list and the route manifest and emits only the entries whose gate the user satisfies, so adding a permission to a role changes the menu on the next login with no front-end deploy. The rejected alternative was a hand-maintained role-to-menu map, which is faster to read and wrong the first time someone edits a role in production.

A screen the operator cannot reach is absent from navigation, not present and disabled. This is a deliberate departure from the usual advice, which is that greying out an item teaches the interface. A disabled "Payout runs" entry tells an OPS_AGENT that payout runs exist, and a disabled "Fleet KYC review" tells them which fleets are mid-onboarding; in a back office that holds three tenants' commercial terms, the existence of a screen is itself information. The cost is real — operators occasionally cannot find a screen a colleague described, and the answer is the effective-permission drawer on AD-40 rather than a support ticket.

Deep-linking is handled separately from navigation, because hiding a menu entry is not access control. A direct GET of a route the user cannot satisfy renders the denied state from section 5.10 and the API returns 403 FORBIDDEN regardless of what the client did. Permission checks live on the server in every case; the front-end array only decides what to draw.

THE ONE ROLE THAT CANNOT BE WIDENED

AUDITOR holds every *.read permission plus audit.read and no write permission at all. It is the seat we hand an external accountant. It still sees last-four-only PII, because there is no pii.read.raw permission anywhere in the Permission union to grant (invariant I-06), and export.create is deliberately not in its set.

5.3 THE FORTY-ONE SCREENS

The complete register, grouped by the operator task it serves

Every one of the 41 screens is listed below with the route it lives at, the permission that gates it and what an operator does on it. The PERMISSION GATE column is the single permission the router checks before rendering; individual actions inside a screen carry their own, usually stricter, gate, and those are named in sections 5.4 to 5.8. The three non-session values in that column — public, challenge and session — are the login chain. AA-11 assigns each of these 41 ids to a build week and an owner; nothing here implies an order of work.

Routes are relative to admin.atavishaala.in. Screens whose route contains a parameter are detail views reached from a list, never typed by hand, and every one of them handles a 404 for an id the operator's tenant scope cannot see identically to a 403 — the denied state in section 5.10 — so that probing ids reveals nothing about what exists.

Sign-in and factor enrolment — AD-01 to AD-03

Three screens the operator sees once a day and once ever. Password verification and TOTP are separate steps because POST /admin/auth/login returns a challenge token, not a session; the session only exists after POST /admin/auth/totp. Invariant I-09 means there is no path past AD-02 for a staff user with no registered factor, which is why AD-03 exists as a forced interstitial rather than a settings page: staff_users.totp_enrolled_at must be non-null before a refresh cookie is issued at all.

ID	ROUTE	SCREEN	PERMISSION GATE	WHAT IT DOES
AD-01	/login	Sign in	public	Email + password. No org selector. Generic error text only.
AD-02	/login/totp	Two-factor challenge	challenge	Six digits, 30 s ring, ten-attempt lockout.
AD-03	/onboarding/totp	Enrol authenticator	session	QR plus manual key, then ten recovery codes shown once.

Dashboard, orders, shipments and returns — AD-04 to AD-09

This block answers "where is this customer's order" and "what is late". AD-07 is where an ops shift actually sits: a board grouped by shipment state with a lane for shipments past their ready_by, because a supplier who has not packed by the promised time is the earliest signal that a delivery will breach. AD-04 is the shift-start glance and nothing more — eleven tiles from one query, each linking into the queue behind it.

AD-06 is the investigation screen and section 5.5 specifies it in full. Note what is not in this block: there is no "edit order" screen, because orders.status is derived by reduceOrderStatus() under invariant I-11 and the shipment is the only thing an operator can act on.

ID	ROUTE	SCREEN	PERMISSION GATE	WHAT IT DOES
AD-04	/	Operations dashboard	order.read	Eleven tiles: SLA breaches, unassigned jobs, moderation backlog, COD exposure.
AD-05	/orders	Order list	order.read	Saved views, 20 filters, masked customer column, CSV export button.
AD-06	/orders/:id	Order detail	order.read	Shipment cards, payment panel, event timeline, internal notes.
AD-07	/shipments	Shipment board	order.read	Kanban by state with a ready_by breach lane. The ops home screen.
AD-08	/shipments/:id	Shipment detail	order.read	Items, supplier, linked job, force-transition drawer.
AD-09	/returns	Return queue	order.return.review	Evidence thumbnails, window countdown, decide drawer.

Delivery operations — AD-10 to AD-14

The exception board AD-11 is the most-used screen in the portal and section 5.4 specifies it separately. The other four support it: AD-10 for the whole job population when someone asks a question the board cannot

answer, AD-12 for the offer-wave and POD evidence behind one job, AD-13 for the POD records the geofence flagged, and AD-14 to answer "do we even deliver to this pincode" without opening a fleet record.

AD-12 shows the wave history with the OFFER_WAVE_TTL_SECONDS clock for each of the three waves (600, 600, 900 seconds), which is what an operator reads to decide between rebroadcasting and assigning directly. The POD gallery shows the captured geo distance against POD_GEOFENCE_METRES = 200 as a number, not a map: Maps tiles are billed and AA-00 section 0.9 names rider GPS tracking as a paid gap, so the distance is computed server-side and rendered as text.

ID	ROUTE	SCREEN	PERMISSION GATE	WHAT IT DOES
AD-10	/jobs	Delivery job list	job.read	Filter by state, fleet, pincode; promised_by breach highlight.
AD-11	/jobs/unassigned	Unassigned exceptions	job.read	Wave history per job, assign and rebroadcast actions inline.
AD-12	/jobs/:id	Job detail	job.read	Offer waves, rider, scan trail, POD gallery with geo distance.
AD-13	/pod-review	POD review	job.pod.review	Outside-fence and duplicate-photo candidates, oldest first.
AD-14	/serviceability	Pincode serviceability	job.read	Search a pincode; see fleets, fees, COD flag, gaps map.

Catalog and moderation — AD-15 to AD-20

CATALOG_MANAGER is the only role that approves a partner listing, and AD-18 is where that happens: a side-by-side diff of the proposed revision against what is live, with approve, reject and delist as the three outcomes of POST /admin/products/:id/moderate. The queue is ordered oldest first with an age column, because a partner waiting four days on a listing is a supplier relationship problem before it is a catalog problem.

AD-20 is the only screen in the portal that writes a number into inventory, and it writes signed deltas with a reason code rather than an absolute count. Setting stock to 40 hides the fact that it was 55 and eleven units cannot be found; posting -15 with a reason code makes the shrinkage countable. AD-17 creates house products only — partner products never originate here, they arrive through review.

ID	ROUTE	SCREEN	PERMISSION GATE	WHAT IT DOES
AD-15	/products	Product list	catalog.read	Partner column, moderation chip, stock band, bulk delist.
AD-16	/products/:id	Product detail	catalog.read	Variants, media, price history, moderation history.
AD-17	/products/new	Create house product	catalog.write	Six-step wizard with a draft autosave.
AD-18	/moderation	Moderation queue	catalog.moderate	Side-by-side diff of proposed against live, approve or reject.
AD-19	/categories	Category tree	catalog.read	Drag to reparent, inline rename, SKU count per node.
AD-20	/inventory	Stock adjustments	catalog.stock.adjust	Signed deltas with reason codes and an audit tail.

Product-company supervision — AD-21 to AD-24

PARTNER_MANAGER owns the supplier relationship end to end, and these four screens are the whole of it: the population with the five numbers that matter (status, live SKUs, 30-day GMV, on-time rate, KYC completeness), the single-company record in six tabs, the invite that creates a DRAFT org, and the KYC document queue. Section 5.6 covers what each action costs in authentication.

AD-22's bank tab shows the account number and IFSC as last four only and a verified_at stamp; the verification itself is a staff member sending ₹1 by UPI and matching the holder name, because penny-drop APIs bill roughly ₹3 per check. That is a paid gap, not a solved problem, and it is the reason partner.bank.verify is step-up gated: the button records a human assertion, so we want to know exactly which human.

ID	ROUTE	SCREEN	PERMISSION GATE	WHAT IT DOES
AD-21	/partners	Product companies	partner.read	Status, live SKUs, 30-day GMV, on-time rate, KYC completeness.
AD-22	/partners/:id	Partner detail	partner.read	Six tabs: profile, users, KYC, rate cards, bank, activity.
AD-23	/partners/invite	Invite a company	partner.invite	Legal name, GSTIN, owner email. Creates a DRAFT org.
AD-24	/partner-kyc	Partner KYC review	partner.kyc.review	Document viewer with verify and reject, note mandatory on reject.

Delivery-company supervision — AD-25 to AD-30

Structurally parallel to the partner block, with three additions. AD-28 adds insurance-expiry validation, because an expired policy on a vehicle carrying a ₹40 000 order is a liability we would rather catch in review than in a claim. AD-29 is the service-area request queue; cut C-8 removes the fleet-side submission workflow, so for these ten weeks an admin edits fleet areas directly and this screen mostly renders empty — it is built because the endpoint and the coverage-delta preview are what make the cut reversible in one sprint.

AD-26's vehicles tab is present in the layout and unpopulated: cut C-7 removes the vehicle registry entirely, so riders exist and vehicles do not, and heavy equipment routes to ATV-SELF. AD-30 is the only cross-fleet view of riders, which is what an operator needs when the same person appears under two fleet orgs with two different licence expiry dates.

ID	ROUTE	SCREEN	PERMISSION GATE	WHAT IT DOES
AD-25	/fleets	Delivery companies	fleet.read	Tier, live riders, acceptance rate, on-time rate, COD exposure.
AD-26	/fleets/:id	Fleet detail	fleet.read	Seven tabs adding riders, vehicles and service areas.
AD-27	/fleets/invite	Invite a fleet	fleet.invite	Legal name, GSTIN, transport licence, owner email.
AD-28	/fleet-kyc	Fleet KYC review	fleet.kyc.review	Adds insurance-expiry validation to the partner flow.
AD-29	/fleet-areas	Service-area requests	fleet.area.approve	Pending pincode and rate changes, with a coverage delta preview.
AD-30	/riders	Rider directory	fleet.read	Cross-fleet rider view with suspension action and licence expiry flags.

The Advisor console — AD-31 to AD-33

Three screens for the AGRONOMIST seat, specified in full in section 5.8. They are separated from the ops blocks above because the operator is a different person with a different clock: a two-hour business-hours SLA rather than a delivery promise, and a queue ordered by `sla_due_at` rather than by event time.

ID	ROUTE	SCREEN	PERMISSION GATE	WHAT IT DOES
AD-31	/advisory	Advisory queue	advisory.read	Ordered by <code>sla_due_at</code> with a live countdown and a breach banner.
AD-32	/advisory/:id	Advisory console	advisory.claim	Media gallery, crop context, claim, note form, callback log.
AD-33	/advisory/shifts	Agronomist shifts	advisory.read	Who is on duty across 08:00–20:00, seven days.

Money — AD-34 to AD-38

The FINANCE seat, specified in section 5.7. Two of these five are read-only by construction: AD-34 renders payments and their settlement state, and the ledger half of AD-36 cannot be edited at all because `wallet_ledger` rows are never UPDATED under invariant I-04. The three that write money — refund approval, payout release, COD reconciliation — are all step-up gated and all idempotent.

ID	ROUTE	SCREEN	PERMISSION GATE	WHAT IT DOES
AD-34	/payments	Payments	payment.read	Razorpay and COD with settlement state and reference last four.
AD-35	/refunds	Refund approvals	refund.approve	Two-person flow: initiator and approver are never the same staff user.
AD-36	/wallets/:userid	Customer wallet	payment.read	Append-only ledger with credit and debit drawers.
AD-37	/payouts	Payout runs	payout.read	Accrued totals, draft assembly, lock, release with UTR capture.
AD-38	/cod	COD reconciliation	cod.reconcile	Expected against declared against received, with variance and age.

Customers, staff and the audit trail — AD-39 to AD-41

AD-39 is a support lookup, not a CRM: masked phone, order count, wallet balance and the return-abuse flag, with suspend and restore behind a step-up. AD-40 carries the effective-permission drawer, which is the answer to every "why can I not see that screen" question, and shows a role diff before a change is submitted so the operator approves a set of permissions rather than a role name.

AD-41 has two tabs and they are not the same register. The audit tab is every mutating admin request with its actor, the permission used, the target and a JSON diff capped at `AUDIT_DIFF_MAX_BYTES = 8192`; it exists because audit is Express middleware on the admin router, not a per-handler call, so a new endpoint cannot forget it. The disclosures tab is the `pii_disclosures` register — who saw which field, when, and under which of the five reasons — and it is the only place in the programme where a PII access shows up at all.

ID	ROUTE	SCREEN	PERMISSION GATE	WHAT IT DOES
AD-39	/customers	Customers	customer.read	Masked phone, order count, wallet, return-abuse flag.
AD-40	/staff	Staff and roles	staff.read	Effective-permission drawer, role diff preview before submit.
AD-41	/audit	Audit and disclosures	audit.read	Two tabs: mutating actions with diffs, and the PII disclosure register.

5.4 THE EXCEPTION BOARD

AD-11, GET /admin/jobs/unassigned, and the twenty-second timer behind it

AD-11 is the screen an ops shift keeps open. Its primary feed is GET /admin/jobs/unassigned — in AA-04 section 4.6 described, accurately, as the screen ops lives on — and the board around it aggregates five other exception sources so that one person watching one tab sees everything that has stopped moving. The alternative was six separate queues reached from the dashboard tiles, which is cleaner to build and loses the operator two minutes per exception in navigation.

Unassigned jobs Three offer waves expired: 600 + 600 + 900 s. Job is UNASSIGNED.	POD disputes Geo outside the 200 m fence, or a reused photo hash. Oldest first.	COD shortfalls Declared minus received beyond COD_VARIANCE_BLOCK_BPS = 200 bps.	KYC rejections A partner or fleet document rejected and not resubmitted.
▼ one board, one poll, one ordering rule			
Advisory breaches sla_due_at in the past with no recorded note. Also on AD-31.	Stuck payments Authorised but not captured, or captured with no order transition.	Ordering Oldest BREACH first, not newest event. Age is measured from the breach.	Claim Atomic for advisory. FSM-conflict for the rest. See below.

The ordering rule is the part that gets implemented wrongly. Sorting by created_at or by the most recent event puts the freshest exception on top, which feels responsive and is exactly backwards: the row that has been broken longest is the row closest to a refund, and a busy shift working from the top of a newest-first list never reaches it. The board sorts ascending by the moment the promise was broken — promised_by for a job, sla_due_at for an advisory thread, the remittance due date for a COD shortfall — and renders that age in the first column so the ordering is visible rather than implied. There is no user-selectable sort on the age column, because a sort control is how the rule gets quietly disabled.

Why it polls every twenty seconds instead of streaming

The board re-fetches every 20 seconds. AA-04 section 4.10 is blunt about the consequence: that timer is the largest single source of Cloud Run requests in the programme. One operator with the tab open costs 180 requests an hour, four operators across a ten-hour shift cost roughly 7 200, and that is before anyone opens a second tab.

Server-sent events would cost almost nothing per update and were rejected twice. Cut C-1 removes the SSE offer stream on the fleet side for two dev-days, and the same reasoning applies here: Cloud Run scales to zero, a held SSE connection is a request that never ends, and a container kept warm by an idle stream is the one thing that turns a Rs 0 bill into a real one. Polling also survives the container being recycled mid-shift, which a stream does not without reconnect logic nobody has time to test in week five.

The escape hatch is named in advance so it is not a debate later: if request count approaches the free-tier threshold, the poll interval goes to 45 seconds before anything else changes. Nothing about the screen breaks at 45 seconds — an exception that has been open eleven minutes is not meaningfully worse at eleven minutes and forty-five seconds — and it is a one-line change to a constant. Two smaller mitigations ship from day one: the poll pauses when the tab is hidden via the visibilitychange event, and the response carries an ETag so an unchanged board returns 304.

Claim before you act

Two operators working the same row is the failure this screen has to prevent, and it is not hypothetical: a Monday morning board with forty rows and three agents will collide within the hour. Rebroadcasting a job someone else has just force-assigned wastes a wave and confuses a fleet.

The advisory queue has a real atomic claim: POST /admin/advisory/:id/claim takes the row in one statement and the second claimer gets 409 ALREADY_CLAIMED. The other five sources have no claim endpoint in AA-04, and rather than invent one we rely on the state machine, which is a weaker but honest guard. Every action on the board is a transition from a specific state, so the second operator's POST fails with 409 CONFLICT once the job has left UNASSIGNED, and the board renders a "worked by" hint on rows whose most recent audit_log actor is

another staff user within the last five minutes. That hint is advisory. It is derived from an audit read, it can be up to 20 seconds stale, and it does not stop anyone — it only makes the collision visible before the click rather than after.

A claim goes stale when an agronomist claims a thread and records nothing. The row stays on the breach list, because breach is computed from `sla_due_at` and not from claim state, and any holder of `advisory.reassign` can move it with `POST /admin/advisory/:id/reassign`, which transfers the claim and keeps the original clock. Keeping the clock matters: resetting it on reassignment would let a thread be laundered clean by passing it between two agronomists.

Row actions

ACTION	ENDPOINT	PERMISSION	STEP-UP	WHAT IT DOES
Assign to a fleet	<code>POST /admin/jobs/:id/assign</code>	<code>job.assign</code>	no	Skips the auction. Requires an Idempotency-Key; a replay returns the first response byte for byte under invariant I-12.
Rebroadcast	<code>POST /admin/jobs/:id/rebroadcast</code>	<code>job.rebroadcast</code>	no	Wave 1, 2, 3 or ALL. Fails 409 if the job is no longer UNASSIGNED.
Force to ATV-SELF	<code>POST /admin/jobs/:id/assign</code>	<code>job.assign</code>	no	The same endpoint with our own fleet org. The last resort, and the reason cut C-7 is survivable.
Cancel the job	<code>POST /admin/jobs/:id/cancel</code>	<code>job.cancel</code>	yes	Returns the shipment to <code>READY_FOR_PICKUP</code> so it can be re-offered. Does not cancel the order.
Flag a POD	<code>POST /admin/pod/:id/flag</code>	<code>job.pod.review</code>	no	Marks the record suspect: geo outside the fence, or a photo hash seen before.
Reconcile COD	<code>POST /admin/cod/:id/reconcile</code>	<code>cod.reconcile</code>	yes	Records the received amount and the variance. Closes the period for that fleet.
Review KYC	<code>POST /admin/partner-kyc/:id/review</code>	<code>partner.kyc.review</code>	no	VERIFY or REJECT with a note. The note is mandatory on reject.
Claim an advisory	<code>POST /admin/advisory/:id/claim</code>	<code>advisory.claim</code>	no	Atomic. The second claimer gets 409 <code>ALREADY_CLAIMED</code> and the board removes the row on its next poll.
Open the detail	<code>GET /admin/jobs/:id</code>	<code>job.read</code>	no	A drawer, not a route change, so the board keeps its scroll position and its poll.

THE ACTION THAT IS NOT HERE

There is no click-to-call. No free provider offers a masked call bridge in India, and exposing a rider or customer number to reach one would break invariant I-06. An operator who has to reach a customer sends a WhatsApp template from the order screen. Masking through Exotel or Kaleyra is roughly ₹0.60/minute plus ₹1 500/month, named as a paid gap in AA-00 section 0.9.

5.5 ORDER AND SHIPMENT INVESTIGATION

AD-06 and AD-08: answering "where is this order" without an editable status field

The most common support question is a customer asking where their order is, and it is harder to answer here than in a single-warehouse shop because one order splits into one shipment per supplier, and each shipment gets its own delivery job, its own offer auction and its own promise. An order can be simultaneously delivered, out for delivery and unassigned. AD-06 is built around that fact rather than hiding it.

1. Search AD-05 by order number, or by the last four digits of the phone the customer reads out. There is no full-number search because there is no full number to search on; the index is an HMAC column, so last-four matching is a suffix query on masked text.
2. Open AD-06. The header shows the derived order status from `reduceOrderStatus()` and, beside it, the shipment count and how many of them are late. The derived value is rendered as text with no control attached to it.
3. Read the shipment cards. Each carries its supplier, its state, its `ready_by` and `promised_by`, and the linked delivery job with that job's current wave. This is where the answer usually is: one supplier has not packed.
4. Open the event timeline, read from `order_events` via `GET /admin/orders/:id/events`. Append-only, newest first, with the actor on every row — a staff email, a fleet org, a rider id or the string system for an FSM-driven transition.
5. If the answer needs the job, open AD-12 from the shipment card. Offer waves, the rider, the scan trail and the POD gallery with its geo distance in metres are all there.
6. Record what you told the customer as an internal note through `POST /admin/orders/:id/notes`. It is staff-visible only and it is the reason the next agent does not repeat the investigation.

THERE IS NO STATUS DROPDOWN, AND THERE NEVER WILL BE

`orders.status` is a projection of its shipments produced by `reduceOrderStatus()`, and invariant I-11 means no endpoint in any namespace writes it. A unit table drives all 28 shipment-state combinations to an expected order status. If an order looks wrong, the shipment underneath it is wrong; fix that and the order follows on the next recompute. A status field an operator can set would make the projection a lie in exactly the cases where someone is already confused.

AN OPERATOR CAN	HOW, AND WHAT IT COSTS
Cancel a shipment before handover	<code>POST /admin/orders/:id/cancel</code> with <code>order.cancel</code> . Releases the reserved stock and reverses the payment or voids the COD line. Idempotency-Key required.
Re-split a shipment	<code>POST /admin/shipments/:id/resplit</code> with <code>order.shipment.reassign</code> . Moves items into a new shipment, which creates a new delivery job and a new promise.
Re-offer a delivery job	<code>POST /admin/jobs/:id/rebroadcast</code> with <code>job.rebroadcast</code> , choosing wave 1, 2, 3 or ALL.
Force-assign to ATV-SELF	<code>POST /admin/jobs/:id/assign</code> with <code>job.assign</code> . Our own fleet, no auction. Used for heavy equipment and for anything nobody accepted.
Force a shipment transition	<code>POST /admin/shipments/:id/transition</code> with <code>order.transition.force</code> . Step-up, and a reason of at least 20 characters that lands in <code>audit_log</code> .
Approve a refund	<code>POST /admin/refunds/:id/approve</code> with <code>refund.approve</code> . Step-up. The initiator and the approver are never the same staff user (AD-35).
Add an internal note	<code>POST /admin/orders/:id/notes</code> with <code>order.note.write</code> . Never shown to the customer.
AN OPERATOR CANNOT	WHY
Edit an order status	Invariant I-11. It is derived. There is no endpoint, no column setter outside <code>orders/fsm.ts</code> , and an ESLint rule forbids <code>{ status: }</code> inside a prisma update outside an fsm file.
Edit a ledger row	Invariant I-04. A Postgres rule raises on <code>UPDATE</code> or <code>DELETE</code> against <code>wallet_ledger</code> , <code>partner_payouts</code> and <code>fleet_payouts</code> . A correction is a compensating row.
See a full phone number	Invariant I-06. A Prisma client extension masks at the ORM boundary, so the value never reaches the serialiser. There is no reveal control and no permission that unmask.
See a full delivery address	AD-06 shows locality and pincode. The one exception in the programme is <code>GET /fleet/jobs/:id/manifest</code> , which is fleet-side, job-scoped, expires after <code>MANIFEST_WINDOW_MINUTES = 240</code> and writes a <code>pii_disclosures</code> row in the same transaction. It releases the address only; the phone is never released.
Cancel a job that is already delivered	<code>assertTransition</code> rejects it with 409. The remedy is a return through AD-09, not a state edit.
Act without leaving a trace	Invariant I-08. Audit is middleware on the admin router, so a mutating request cannot avoid writing a row with actor, permission, target and a diff capped at 8 192 bytes.

5.6 PARTNER AND FLEET SUPERVISION

Onboarding, KYC, service areas and the switches that change who gets offered work

A product company and a delivery company follow the same lifecycle — invited, DRAFT, documents submitted, verified, ACTIVE, and possibly suspended — so AD-21 to AD-24 and AD-25 to AD-30 are the same screens twice with different fields. That is deliberate repetition rather than a shared generic component: the KYC document set differs, the fleet flow adds insurance-expiry validation, and a single abstracted "organisation" screen with two conditional branches would be harder to read than two honest copies. AA-08 section 8.2 carries the full permission list; what follows is what the screens do with it.

An invite creates a DRAFT org and emails the first owner user through Brevo. Nothing about that org is live: it has no SKUs, it receives no offers, and its users can sign in only to complete their own onboarding. That is why the invite is not step-up gated — it grants no commercial capability on its own, and the gates sit later, on bank verification and on status changes.

KYC review is a document viewer with verify and reject, and a note that is mandatory on reject because the rejection text is what the supplier sees and what a second reviewer reads three weeks later. Opening a document image writes a pii_disclosures row under reason KYC_REVIEW — the reviewer sees the image, which is unavoidable, and the register records that they did. The extracted numbers beside the image are still last four only.

ACTION	PERMISSION	FRESH TOTP	NOTE
Invite a product company	partner.invite	no	Legal name, GSTIN, owner email. Creates a DRAFT org (AD-23).
Invite a delivery company	fleet.invite	no	Adds the transport licence number. Creates a DRAFT fleet org (AD-27).
Verify or reject KYC	partner.kyc.review / fleet.kyc.review	no	Note mandatory on reject. Fleet adds insurance-expiry validation (AD-24, AD-28).
Verify a bank account	partner.bank.verify	yes	Records a human assertion after a manual ₹1 UPI check. Penny-drop is a paid gap at about ₹3.
Suspend or terminate a partner	partner.status.write	yes	Suspension delists every SKU the company has live. Reversible; the delisting is not automatic on restore.
Suspend or terminate a fleet	fleet.status.write	yes	Withdraws live offers immediately. Accepted jobs stay with the fleet and have to be cancelled individually.
Change a fleet tier	fleet.tier.write	no	Tier 1 to 3 sets which broadcast wave the fleet sits in. The offer-eligibility switch in practice, and the one a fleet manager reaches for weekly.
Approve a service area	fleet.area.approve	no	Pincode and rate changes with a coverage-delta preview (AD-29). Cut C-8 means an admin usually edits areas directly instead.
Suspend a rider	rider.suspend	no	Reason code mandatory. Reassigns that rider's live jobs, which can put them back on AD-11.

The fresh-TOTP column is read from the permission matrix, not decided here, and it is worth noticing where it says no. Rejecting a KYC document is not step-up gated, because a wrong rejection is recoverable in one click and gating it would slow the queue an operator works through thirty times a day. Verifying a bank account is gated, because the next thing that happens to that account is a payout release. The rule the matrix encodes is that step-up follows irreversibility and money, not sensitivity in the abstract. `requireFreshFactor(300)` guards twenty-eight endpoints in total under invariant I-09; AA-08 section 8.4 lists them.

Suspension asymmetry is the trap in this block. Suspending a fleet withdraws live offers instantly, but jobs it has already accepted stay with it, so a fleet manager who suspends a fleet at 14:00 still has to walk its accepted jobs off one at a time through AD-12. Suspending a partner delists its SKUs, and restoring the partner does not relist them, because the alternative — mass relisting on restore — would put stale prices and stale stock back on the shelf in one write.

5.7 MONEY SCREENS

AD-34 to AD-38: read-only by construction, two-person where it pays out

The FINANCE seat cannot touch catalog or orders, and the five money screens are built on the assumption that the people using them are careful, tired and occasionally wrong. Three structural choices carry most of the safety: the ledger cannot be edited at all, a refund needs two different staff users, and a payout run has to be assembled and looked at before it can be released.

AD-36 renders `wallet_ledger` as an append-only list, newest first. There is no edit control on a row because invariant I-04 means a Postgres rule raises on UPDATE and DELETE against the three ledger tables; the screen is read-only because the database is. A correction is a compensating entry through `wallet.credit` or `wallet.debit`, both step-up gated, and the viewer groups a compensating pair under the entry it reverses with the net shown once, so an operator reading a balance does not have to do the arithmetic that produces a wrong refund. The debit endpoint refuses to take a balance negative. Cash-out does not exist anywhere in the programme: wallet value is spendable on Atavishaala and nothing else, because paying it back to a bank account is prepaid-instrument business that needs an RBI licence.

AD-35 is the two-person refund. POST `/admin/refunds` with `refund.initiate` creates a `PENDING_APPROVAL` row and pays nothing; POST `/admin/refunds/:id/approve` with `refund.approve`, behind a fresh TOTP, chooses GATEWAY or WALLET and releases it. The server rejects an approval whose actor equals the initiator, so the rule is enforced in the module and not in the button state. With a small team this is genuinely inconvenient — on a Sunday there may be one person online — and the answer is that the refund waits, because a single-actor refund path is the first thing an auditor asks about and the easiest thing to abuse.

1 Accrued GET <code>/admin/payouts/accrued</code> . What the next run would contain, by org and kind.	2 Draft POST <code>/admin/payout-runs</code> . Assembles a DRAFT. Withhold lines here, with a reason.	3 Lock POST <code>.../lock</code> . Freezes membership, generates the bank file. Still pays nothing.	4 Release POST <code>.../release</code> . Step-up. Records the UTR. Irreversible.
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The lock step exists so that release is boring. Locking freezes run membership and generates the bank file; the screen then shows the dry run — line count, gross, withheld, net, and the per-org breakdown that the bank file will produce — and the release control stays disabled until the operator has actually opened that view. It is a soft gate, recorded in the audit diff rather than enforced by the endpoint, and it is worth the one extra click because release is the only action in the portal with no compensating action behind it: once money has left the account, the remedy is a phone call to a bank.

Release is idempotent on the run key. A double-click, a retried request or a browser that reconnects mid-POST replays the first response byte for byte under invariant I-12, because the Idempotency-Key is unique on (actor_id, endpoint, key) and the run id is the key. This is the one place in the portal where the idempotency contract is load-bearing rather than tidy: without it, the same batch can be released twice and the second release looks identical to the first in every log we keep.

AD-38 reconciles COD. Expected against declared against received, with the variance in basis points and the age of the open remittance. A fleet past `COD_VARIANCE_BLOCK_BPS = 200` — two per cent — is blocked from further COD work, and unblocking is a deliberate act with a mandatory reason that lands in the audit diff. The reason field is not politeness: a fleet unblocked three times in a month with no recorded reason is a pattern nobody can see afterwards.

MONEY FORMATTING, AND WHY NO SCREEN ACCEPTS RUPEES

Every amount in the portal is rendered by the kit Money component from an integer paise value on a `BIGINT*_paice` column, and every amount is entered the same way. There is no rupee-decimal input anywhere in the admin portal: an operator types into a paise-safe field that shows the rupee value it will send and submits an integer. Accepting "1234.5" means a float somewhere, and a float in a refund is a rounding error the ledger records permanently. The component also renders the currency once per row rather than per cell, so a column of amounts stays right-aligned and comparable at a glance.

5.8 THE ADVISOR CONSOLE

AD-31 to AD-33, the AGRONOMIST seat, and a two-hour clock that stops at 20:00

Crop Advisor is the part of the product a farmer judges us on, and its promise is a response within two business hours. The console is three screens: the queue, the request, and the shift roster. The AGRONOMIST role holds all Advisory permissions except `advisory.sla.override`, plus `customer.read`, and nothing else — no orders, no catalog, no money. An agronomist is a domain expert we may not employ directly, and the permission set is drawn on that assumption.

The clock is business-hours, not wall-clock. The window is 08:00 to 20:00 IST, seven days a week, and `sla_due_at` is computed once at insert by the advisory module (task P8-BE1-02 in AA-11): a question submitted at 09:15 is due at 11:15, and one submitted at 19:30 is due at 09:30 the next morning because the two hours pause overnight. Computing it once and storing it is the whole design decision. A due time derived on read would move if the window definition changed, if a server's timezone were wrong, or if a row were read on a day the roster differed, and a promise that silently moves is worse than no promise. The stored value also lets the queue be a plain ORDER BY `sla_due_at` ASC, which is a b-tree index scan rather than a computation per row.

AD-31 is that ordering with a live countdown per row and a breach count in the header from GET `/admin/advisory/queue`. Breached threads stay at the top and are not hidden, archived or auto-closed; the list of things we have already failed at is short, visible, and the first thing an agronomist starting a shift sees. Only `advisory.sla.override` can move a due time, it is step-up gated, it requires a written reason of at least 20 characters, and the AGRONOMIST role deliberately does not hold it — the person whose SLA it is cannot extend their own clock.

1. Claim from the queue. POST `/admin/advisory/:id/claim` is atomic; the second claimer gets 409 ALREADY_CLAIMED and their board drops the row on its next poll.
2. Read the request on AD-32: the farmer's description, the crop, the season, the pincode, and the photo gallery from Cloudinary at a size worth looking at.
3. Reply to the farmer. The console renders the last four digits of the phone and a WhatsApp template send; there is no tel: link and no dial control, because the full number is not present in the response to render one from.
4. Record the outcome through POST `/admin/advisory/:id/notes`: diagnosis, recommendation, call duration. Recording the note is what stops the clock, not claiming the row.
5. If it is not yours to answer, POST `/admin/advisory/:id/reassign` moves the claim and keeps the original `sla_due_at`.

THE GAP IN THE MIDDLE OF THIS FLOW

The console has a callback log and the note form asks for a call duration, but the portal cannot place the call. Masked calling is a paid gap: roughly ₹0.60/minute plus a ₹1 500/month platform fee with Exotel or Kaleyra. Until that is bought, a voice conversation happens because the farmer rings the published support line after the WhatsApp reply, and the agronomist types the duration in by hand. It is a real hole and it is on the paid-gap list in AA-00 section 0.9, not presented as solved.

WHAT AN AGRONOMIST NEVER SEES

The question, the photos, the crop, the season and the pincode — yes. The farmer's phone number, their address, their order history and their wallet — no. `customer.read` grants a masked profile only, and there is no permission in the union that unmask it for anyone, under invariant I-06.

AD-33 is the shift roster: who is on duty across the 08:00 to 20:00 window, seven days. It is a read screen with no scheduling engine behind it, which is the right amount of tool for a rota of two or three people, and it exists mainly so that a breach at 18:40 on a Sunday has a name attached to it. Task P8-FE1-01 in AA-11 fixes the queue's minimum working width at 1366 px, because the machine an agronomist actually uses is a borrowed laptop and not the 1920 px monitor the rest of the portal is designed against.

5.9 INTERACTION AND ACCESSIBILITY STANDARDS

A queue tool is a keyboard tool, and this one is used eight hours a day

The people who live in this portal are not visiting it. An operations associate works the exception board for a full shift, and the difference between a two-second row action and a six-second one is roughly an hour of their day. That single fact sets the interaction standard: every action on a queue screen must be reachable from the keyboard without leaving the row, and the mouse is an alternative rather than the primary path.

Concretely, the row list owns focus. Up and down move the selection, Enter opens the drawer, Escape closes it and returns focus to the row that was open, and the primary action of each queue has a single-key shortcut shown in the row on hover and in a "?" overlay. There is no interaction anywhere in the portal whose only path is drag-and-drop, because a drag target cannot be reached by a keyboard and cannot be described to a screen reader. Where an order is expressed by dragging in some other product, this one uses an explicit position field.

AREA	THE STANDARD	HOW IT IS CHECKED
Focus	Every interactive element has a visible focus ring at 3:1 contrast against its background. The drawer and the step-up dialog trap focus and restore it on close.	axe-core rule set in CI plus a manual tab-through of the twelve highest-traffic screens.
Colour	Colour is never the only carrier of state. A blocked fleet reads BLOCKED in text, an SLA breach carries a label, and a variance is signed as well as coloured.	A greyscale screenshot review at the end of each week. If a state disappears, it fails.
Contrast	WCAG 2.1 AA: 4.5:1 for body text, 3:1 for large text and for the boundary of any control. The palette in the kit is fixed at these ratios so a screen cannot drift.	Contrast is computed from the kit tokens in a unit test, not eyeballed per screen.
Tables	Real table semantics: th with scope, a caption naming the queue, and aria-sort on the sorted column. A div grid is not acceptable for tabular data.	CI gate G-09 fails the build on an axe violation of serious or critical severity.
Async results	A polite live region announces the outcome of every mutation: "Job JOB-4412 re-offered to three fleets", not a toast that a screen reader never sees.	Reviewed per pull request against the four-state checklist in section 5.10.
Zoom and width	Usable at 1366 px and at 200 % browser zoom with no horizontal scroll on any screen except the ledger viewer, which scrolls its table body only.	Two fixed viewport sizes in the manual review, 1366 and 1920.

THE PART THAT WILL BE HARDEST, SAID PLAINLY

Focus management around the step-up dialog. A step-up interrupts a mutation the operator has already committed to: they press Release payout, the request comes back 401 STEP_UP_REQUIRED, a TOTP dialog appears, and after a correct code the original request must be retried with the same Idempotency-Key and focus must land where it started. Every part of that is easy to get subtly wrong — the retry silently dropping, the key being regenerated and producing a second payout, focus jumping to the top of the page, or the dialog appearing while another poll is mid-flight and stealing focus mid-keystroke. It lives once in the kit StepUpDialog and Idem hook for exactly this reason (AA-01 section 1.5), and it is the one component with a mandatory second reviewer.

5.10 EMPTY, LOADING, ERROR AND DENIED

The four states every screen implements before it is considered built

A screen is not done when the happy path renders. Definition-of-done item D-11 in AA-00 section 0.16 requires all four states, and they are reviewed as part of the pull request rather than filed as follow-up work. The reason is narrow and practical: a queue tool spends a lot of its life empty, because an empty exception board is the goal state. If empty renders as a blank rectangle, the operator cannot tell it apart from a failed fetch, and the first thing they do is reload.

STATE	WHAT IT MUST DO	WHAT IS FORBIDDEN
Empty	Name the queue, say why it is empty in operator language ("No unassigned jobs. Escalations appear here after the third offer wave expires."), and show the timestamp of the last successful poll so the emptiness is trustworthy.	A bare illustration with no timestamp. A call to action that the role cannot perform.
Loading	Skeleton rows at the real row height on first load only. Subsequent polls update in place and never blank the table — a board that flickers every 20 seconds is unusable.	A full-page spinner on a poll. Any layout shift once data arrives.

STATE	WHAT IT MUST DO	WHAT IS FORBIDDEN
Error	Map the error code from the envelope to copy that names the next action, keep the last good data visible behind a banner, and offer a retry that reuses the same request.	Rendering the raw code or the server message. Discarding rows the operator was reading.
Denied	A 403 PERMISSION_DENIED on a screen reached by a stale link states which permission is missing by name and who grants it. A 401 ACTOR_MISMATCH sends the user to login.	A generic "Something went wrong". Navigation that still lists a screen the user cannot open.

The code-to-copy map is owned by the portal, not the API. The server returns a stable code string from the contract package and a message intended for a developer; the portal decides what an operator reads. That split is what lets the copy for OFFER_ALREADY_TAKEN differ between the admin portal ("Another fleet accepted this job at 14:32") and the fleet portal ("This job is no longer available"), from one server response, without the server knowing who is asking.

One rule about copy is worth stating because it is broken by default. An operator-facing error names the next action, not the failure. "Could not release the payout" is a description of what happened; "Payout not released - the run is stale. Reload the run and check the dry-run total before releasing" tells somebody what to do at 18:40 on a Friday. The kit ships the mapping as data rather than as a switch statement, so the whole set of operator-facing strings can be read in one file during review, which is the only practical way to keep the tone consistent across forty-one screens built by one person over ten weeks.

The denied state deserves one further note, because it is the one place this portal is deliberately more talkative than the other two. Naming the missing permission on a 403 tells an authenticated staff member that a capability exists which they do not hold. That is acceptable inside a staff console where every session is a named employee with a registered TOTP factor, and it saves a support conversation every time somebody follows a colleague link. It is not acceptable in the partner or fleet portals, where the same situation returns a plain 404: a fleet operator is not an employee, and the existence of a capability is commercial information about how the platform is run.

5.11 WHAT THE ADMIN PORTAL WILL NOT DO IN THESE TEN WEEKS

Seven omissions, each with the workaround an operator actually uses

The list below is not a wish list. Every row is a decision already taken, with the manual procedure that replaces it and the point at which that procedure stops being acceptable. Two of these are money problems rather than time problems, and they appear in the paid-gap register in AA-09 with real prices; the rest are on the cut list and are recoverable in dev-days.

NOT BUILT	WHY NOT	WHAT AN OPERATOR DOES INSTEAD	WHEN THIS BREAKS
Click-to-call a customer or rider	Number masking has no free provider in India. Paid fix is roughly ₹0.60/minute plus ₹1 500/month.	The portal triggers a WhatsApp template send. Voice happens when the customer rings the published support line, and the operator logs the duration by hand.	Immediately for a dispute that needs a conversation. This is the weakest point in the design.
Saved views and rich export (C-2)	Three dev-days, and fixed default filters cover the pilot queues.	Each queue ships with one opinionated default filter. Anything deeper is a masked CSV download.	Within about a month of live operation. Strongest Phase 2 candidate.
Bulk edit or bulk action on a queue	Every mutation is idempotent and audited per row; a bulk path would need its own idempotency and its own audit shape, and it multiplies the blast radius of a mis-click.	Rows are actioned one at a time. Force-assignment to ATV-SELF is the only action an operator repeats often, and it is one keystroke per row.	Past roughly forty escalations a day, which the pilot will not reach.
A role editor and custom roles (C-4)	Four dev-days. The nine system roles cover every seat that can be named today.	A permission change is a pull request against the role map, reviewed and deployed. It is slower and it is also auditable, which is not a bad trade at this size.	The first time a partner-facing seat needs a permission split we did not anticipate.
Service-area approval workflow (C-8)	Three dev-days. With three fleets a workflow is ceremony.	An admin edits the fleet service areas directly, with the change written to the audit log.	Past about ten fleets, or the first time a fleet disputes a pincode assignment.
In-app notifications and a live stream	No server-sent events or websockets in this programme (AA-04 section 4.10); the board polls every 20 seconds and each poll is a Cloud Run request against the free tier.	The operator keeps the board open. Genuinely urgent events also fire a WhatsApp template to the ops number, which is the only push path that costs nothing.	When an SLA needs sub-20-second reaction. Nothing in the current scope does.
A reporting or BI layer	Metabase Cloud is about \$85/month; a self-hosted instance needs CPU the free tier does not have.	Eleven fixed dashboard tiles from one cached query, plus masked CSV export for anything else. Analysis happens in a spreadsheet, off-platform.	When someone asks a question the eleven tiles cannot answer twice in the same week.

WHAT IS NOT ON THIS LIST, AND WILL NOT BE

Nothing here is a reduction in the PII posture, and no cut buys speed by widening what an operator can see. Invariant I-06 is not on the cut list and cannot be traded for dev-days: there is no reveal control, no full phone number, and no permission in the union that un.masks one. If a queue is slower to work because the operator cannot read a phone number, the queue stays slower.

HOW TO ARGUE WITH THIS SECTION

Bring the operator-minutes. A cut is reversible in the dev-days named in AA-10 and the argument that wins is a measured cost: how many times a day the manual procedure runs, how long it takes, and who does it. AA-00 section 0.17 sets out the process. An argument that a feature is standard in other back offices is not a cost and will not move anything.

Read together with AA-06 and AA-07, the shape of the whole back office should now be visible: the admin portal decides, the supplier portal fulfils, and the fleet portal moves goods, and each one holds only the screens its own operator needs. The forty-one screens above are enumerated as build tasks with owners and checkable outputs in AA-11, and the permission gate on every one of them is defined once in AA-08 rather than restated here.

5.12 THE ELEVEN DASHBOARD TILES, DEFINED

AD-04 is one SQL query, cached thirty seconds, and every tile is a link

AD-04 earns its place only if it answers the question an operator has at 09:00 — what is wrong right now — in one glance and then gets out of the way. It is defined as exactly eleven tiles because that is what one SQL statement can compute without a sequential scan, and because a dashboard that grows a twelfth tile every month becomes a wall nobody reads. Task P4-BE1-03 in AA-11 owns the query, and the response is cached for thirty seconds at the Cloudflare edge, which matters: with four operators and a 20-second board poll, an uncached dashboard would be a meaningful share of the Cloud Run free request budget on its own.

Every tile is a link to a pre-filtered list rather than a drill-down modal. A tile that shows a number the operator cannot act on is a report, and this is not a reporting tool; if the number is non-zero there must be exactly one click between seeing it and working it.

#	TILE	WHAT IT COUNTS	LINKS TO	AMBER / RED BAND
1	SLA breaches	Advisory threads past <code>sla_due_at</code> and still open.	AD-31, breached filter	Any breach is red. There is no amber band on this tile.
2	Advisory due within the hour	Open threads with <code>sla_due_at</code> inside 60 minutes.	AD-31	Amber above two, red above five.
3	Unassigned jobs	Delivery jobs in UNASSIGNED after the third wave expired.	AD-21	Amber at one, red above three.
4	Offers in flight	Open offer waves, and the age of the oldest wave.	AD-22	Red if the oldest wave exceeds its TTL plus 120 s, which means the cron is late.
5	Moderation backlog	Catalog listings pending review, oldest first.	AD-11	Amber above ten, red if anything is older than 24 h.
6	POD disputes	Delivered jobs with an open dispute or a failed OTP escalation.	AD-24	Any open dispute is amber; older than 48 h is red.
7	COD exposure	Cash collected and not yet deposited, summed across fleets, in paise.	AD-27	Banded on each fleet credit limit, not on one global number.
8	COD variance	Remittances in SHORT, and any fleet above <code>COD_VARIANCE_BLOCK_BPS</code> .	AD-27	A fleet above 200 bps is red and is already blocked from new offers.
9	KYC queue	Partner and fleet organisations awaiting document review.	AD-15, AD-18	Amber above three, red if anything is older than 72 h.
10	Refunds awaiting approval	Refund requests raised and not yet released.	AD-25	Amber above two, red if anything is older than 24 h.
11	Payment exceptions	Orders with a captured payment and no shipment, or the reverse.	AD-06, exception filter	Any row is red. This tile is expected to read zero.

WHY THE THRESHOLDS LIVE IN THE PORTAL AND NOT IN A SETTINGS TABLE

The amber and red bands above are portal constants. A threshold in a settings table invites someone to widen it at 18:00 on a bad day so the dashboard goes green, which converts a signal into a comfort. Changing a band is a pull request, it is visible in review, and it is deliberately slower than the incident it would be hiding. The single exception is the per-fleet COD credit limit behind tile 7, which is commercial data and belongs on the fleet organisation row.

5.13 WHEN A SCREEN IS DONE

The checklist a reviewer runs before an admin screen is merged

Forty-one screens across ten weeks is roughly one screen a day for FE-1, and at that rate the only thing that keeps quality even is a checklist that does not move. The list below is applied per screen in review, not per sprint. It is short on purpose: every item is either mechanically checkable or answerable in one sentence, and nothing on it is a matter of taste.

#	THE CHECK	HOW A REVIEWER CONFIRMS IT
1	The permission gate matches AA-08 exactly, and the screen is absent from navigation when the permission is absent — not present and disabled.	Log in as the narrowest role that should not see it and confirm the nav item is gone and the direct URL returns the denied state from section 5.10.
2	All four states render: empty with a poll timestamp, first-load skeleton, error with the last good data behind a banner, denied naming the missing permission.	Four screenshots in the pull request. A missing state is a blocked review, not a follow-up.
3	No PII beyond four characters appears anywhere, including in a tooltip, a page title, a CSV column or a Sentry breadcrumb.	The masked component is the only path to a phone, GSTIN, PAN or bank field, and the pre-commit ten-digit grep passed. Reviewer greps the diff for tel: and for raw field names.
4	Every amount goes through the kit Money component from an integer paise value, and no input accepts a decimal.	Grep the diff for toFixed, parseFloat and Number(on a money path. Any hit fails.
5	Every mutation the screen issues is either read-only or carries an Idempotency-Key from the kit Idem hook, and a step-up gated action retries the original request after the challenge.	Double-submit the action in the browser and confirm one row of effect and one audit entry.
6	A poll updates in place. No blanking, no layout shift, no focus loss while the operator is mid-keystroke.	Open the screen, focus a row, and wait through two poll cycles without touching anything.
7	Keyboard-only operation completes the screen primary task, and axe-core reports no serious or critical violation.	Unplug the mouse. Gate G-09 covers the axe half automatically.
8	Usable at 1366 px and at 200 % zoom with no horizontal page scroll.	Two viewport sizes in the browser. The ledger viewer is the only permitted exception.
9	Nothing on the screen writes orders.status, edits a ledger row, or offers an unmask control.	These three are structural and a reviewer who finds one stops the review and raises it as an invariant breach under AA-00 section 0.17.

THE THREE CHECKS THAT ARE NEVER WAIVED

Items 3, 4 and 9 map directly onto invariants I-06, I-04 and I-11, and none of them is negotiable against a deadline. The other six can be deferred with a named owner and a date when a week is genuinely lost; these three cannot, because each one is cheap to honour while a screen is being written and expensive to retrofit across forty-one of them.

WHAT THIS CHECKLIST DOES NOT CATCH

It does not catch a screen that is technically correct and operationally useless — the right data, the right gate, and still four clicks to do the thing an operator does two hundred times a day. Nothing mechanical finds that. The only defence is that an engineer sits with an operations associate for one shift in week 7, which is task P7-FE1-04 in AA-11 and is easy to skip when the week is tight. Do not skip it.